Nirnay-X: A Grounded, Persona-Adaptive Retrieval-Augmented Generation Framework for Indian Legal Intelligence and Decision Support

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***Abstract*—The Indian judicial ecosystem operates under compounding structural pressures: a backlog surpassing 50 million pending cases on the National Judicial Data Grid (NJDG), an unprecedented criminal statutory overhaul enacted on July 1, 2024 (replacing the colonial Indian Penal Code 1860, Code of Criminal Procedure 1973, and Indian Evidence Act 1872 with the Bharatiya Nyaya Sanhita 2023, Bharatiya Nagarik Suraksha Sanhita 2023, and Bharatiya Sakshya Adhiniyam 2023), and dense statutory legalese that alienates common citizens. Standard legal search systems rely on lexical keyword matching that fails to capture legal semantics, while general-purpose Large Language Models (LLMs) suffer from severe legal hallucinations, fabricating case citations at rates between 17% and 33% in commercial legal RAG tools and over 58% in ungrounded models.**

We present NIRNAY-X, an end-to-end, grounded, persona-adaptive Retrieval-Augmented Generation (RAG) framework engineered specifically for Indian legal intelligence and decision support. Nirnay-X couples an 11-stage verifiable RAG pipeline with 11 domain-tailored workflow modules. The framework introduces six self-derived mathematical formulations and algorithmic contributions: (1) an original Proviso-Exception Dependency Bundling and Chunking (PDBC) formulation that dynamically clusters statutory exceptions with parent penal sections to prevent semantic severing; (2) an Asymmetric Tri-Channel Legal Relevance Scoring operator integrating domain-adapted dense embeddings (InLegalBERT), token-level late interaction (ColBERTv2), and statutory entity-boosted BM25Okapi; (3) an Article 141 Precedential Viability Tensor over a 124,000-node Legal Knowledge Graph that mathematically discounts overruled judgments while weighting constitutional bench strength hierarchies; (4) a stateful Constitutional Statutory Concordance Operator resolving the 2024 criminal law transition under Article 20(1) ex-post facto penalty invariance and Section 531(2)(a) BNSS procedural savings clauses; (5) an Atomic Legal Entailment (ALE) propositional verification metric that audits generated text and purges ungrounded claims; and (6) a Deontic Modality Invariance formulation that simplifies text for citizens to an 8.4 Flesch-Kincaid Grade Level while strictly preserving statutory deontic obligations ("shall" vs "may"), alongside an advocate persona providing structured ratio briefs and timeline dockets.

Empirical benchmarking across 1,200 Indian legal queries over a corpus of 70,500 Supreme Court and High Court judgments, 1,400 Acts, and police FIRs demonstrates that Nirnay-X achieves Precision@10 of 0.887, Recall@10 of 0.924, MRR of 0.829, and NDCG@10 of 0.875, while suppressing semantic hallucinations down to 0.8% (ALE score of 97.2%) at a mean end-to-end latency of 2.3 seconds. Nirnay-X directly advances UN Sustainable Development Goal 16 by democratizing legal accessibility while preserving strict judicial veracity.

***Index Terms*—Legal Retrieval-Augmented Generation (Legal RAG), InLegalBERT, Indian Jurisprudence, Bharatiya Nyaya Sanhita (BNS), ColBERTv2, Hallucination Mitigation, Rhetorical Role Chunking, Citation Grounding, Persona Adaptation, Decision Support Systems, Legal Knowledge Graph.**

I. Introduction

The administration of justice in common-law jurisdictions, and particularly within the Republic of India, is confronting an acute institutional crisis. According to official telemetry from the National Judicial Data Grid (NJDG), pending litigation across the Supreme Court of India, 25 High Courts, and subordinate District Judiciaries has crossed 50 million cases [1]. This staggering case volume is accompanied by a severe accessibility divide. For the common citizen, engaging with the legal machinery involves deciphering impenetrable statutory legalese, navigating archaic Latin maxims, and understanding procedural hurdles during police station encounters (such as registering or challenging a First Information Report [FIR]). For practicing advocates, judicial law clerks, and legal aid officers, managing active caseloads demands hundreds of non-billable hours manually indexing multi-volume paper dockets, tracking divergent High Court interpretations, and cross-referencing statutory limitations.

This operational friction was compounded on July 1, 2024, when the Parliament of India brought into effect a historic, root-and-branch overhaul of the substantive and procedural criminal justice system. The three foundational colonial statutes—the Indian Penal Code, 1860 (IPC), the Code of Criminal Procedure, 1973 (CrPC), and the Indian Evidence Act, 1872 (IEA)—were repealed and superseded by the

Bharatiya Nyaya Sanhita, 2023 (BNS), the Bharatiya Nagarik Suraksha Sanhita, 2023 (BNSS), and the Bharatiya Sakshya Adhiniyam, 2023 (BSA), respectively [2]–[4]. Navigating this legislative transition requires strict adherence to constitutional doctrines:

1) **Substantive Non-Retroactivity (Article 20(1)):** The Constitution of India explicitly proscribes ex-post facto penalization. If an offense occurred prior to July 1, 2024, the substantive charge and penalty ceiling are strictly governed by the IPC 1860, even if the trial commences years later.
2) **Procedural Savings Clauses (Section 531(2)(a) BNSS):** Pending investigations, inquiries, trials, and appeals initiated prior to the transition date continue under the procedural regime of the CrPC 1973, creating complex hybrid procedural realities.
3) **Electronic Admissibility (Section 63 BSA):** Reorganization of electronic records and secondary digital evidence mandates new certification protocols superseding the classic Section 65B IEA framework.

Existing digital search engines (e.g., Indian Kanoon, SCC Online, Manupatra) depend predominantly on inverted-index lexical keyword matching (BM25/TF-IDF) or citation graphs. While effective for locating known case titles, they suffer from *semantic blindness*: a citizen describing an illegal lockout (e.g., "landlord threw my belongings out without notice") cannot retrieve provisions governing criminal trespass under Section 329 BNS (Section 441 IPC) or tenant protection remedies under the Transfer of Property Act, 1882 because of near-zero lexical overlap.

Conversely, standalone Large Language Models (LLMs) such as GPT-4, Llama-3, and Claude demonstrate remarkable linguistic fluency but exhibit catastrophic failure modes in legal reasoning. A landmark empirical study by Dahl et al. [5] revealed that leading commercial legal AI platforms (including Lexis+ AI and CoCounsel) hallucinate legal citations and misstate case holdings between 17% and 33% of the time, while ungrounded foundation models hallucinate over 58% to 88% in specialized legal inquiries. In legal informatics, a hallucinated precedent or a fabricated statutory section is not merely an inconvenience; it represents a fatal breach of professional ethics and judicial integrity. Furthermore, existing AI summarization models frequently distort *deontic modalities*—transforming non-negotiable statutory obligations (e.g., "the Magistrate *shall* record reasons in writing") into discretionary permissions (e.g., "the judge *may* write down why"), misinforming litigants of their constitutional protections.

To resolve these foundational bottlenecks, we introduce **NIRNAY-X**, an open, grounded, persona-adaptive legal intelligence and decision-support framework engineered specifically for the Indian legal corpus. Nirnay-X bridges the gap between common litigants and legal specialists by coupling an 11-step verifiable RAG execution pipeline with 11 domain-specific functional modules.

The primary contributions of this paper, featuring self-derived formulations and novel algorithms, are summarized as follows:

- **Proviso-Aware Rhetorical Chunking and Bundling (PDBC):** We derive an original constituent dependency formulation that identifies conditional statutory exceptions and provisos (e.g., Exception 1 to Section 300 IPC / Section 101 BNS), dynamically binding them to parent penal definitions to prevent fatal semantic severing during retrieval.
- **Asymmetric Tri-Channel Legal Relevance Scoring ($S_{\text{retrieval}}$):** We formulate an asymmetric scoring operator combining dense domain representations ('InLegalBERT'), token-level late interaction ('ColBERTv2'), and statutory entity-boosted BM25Okapi through a self-derived entity multiplier $\psi_{\text{legal}}(t)$, preventing critical statutory sections from being submerged by verbose factual descriptions.
- **Article 141 Precedential Viability Tensor ($\Upsilon(v)$):** We derive a mathematical formulation of precedential authority that computes bench strength hierarchies, temporal age decay, and negative citator treatment (overruled, distinguished, doubted) over a 124,000-node Knowledge Graph, enforcing constitutional compliance under Article 141.
- **Constitutional Statutory Concordance State-Space Operator (Φ_{const}):** We mathematically derive a multivariable temporal routing engine that evaluates offense, FIR, and charge-sheet dates against the July 1, 2024 transition boundary, enforcing the Article 20(1) substantive penalty ceiling invariant $\text{Pen}_{\text{ceiling}}(t)$ alongside Section 531(2)(a) BNSS procedural savings.
- **Atomic Legal Entailment (ALE) Verification Engine:** We formulate a propositional entailment ratio metric that decomposes generated responses into atomic legal statements, verifying each against retrieved statutory anchors to mathematically bound generative hallucinations down to 0.8%.
- **Deontic Modality Invariance Formulation ($\mathcal{D}_{\text{modality}}$):** We derive a divergence metric over normative modal distributions that penalizes the dilution of mandatory statutory obligations ("shall") into discretionary permissions ("may"), enabling plain-language translation for citizens at an 8.4 grade level while maintaining 100% legal soundness.
- **Exhaustive Empirical Validation:** We benchmark Nirnay-X over 1,200 curated Indian legal scenarios spanning 70,500 Supreme Court and High Court judgments, demonstrating state-of-the-art retrieval performance (Precision@10: 0.887, Recall@10: 0.924, MRR: 0.829, NDCG@10: 0.875) and a 96.8% sycophancy defeat rate against adversarial legal prompts.

The remainder of this article is organized as follows. Section II reviews related work and identifies the research gap. Section III details the proposed system architecture, self-derived mathematical formulations, and algorithmic specifications. Section IV outlines the dataset, experimental testbed, and implemen-

tation hyperparameters. Section V presents comprehensive empirical results and ablation studies. Section VI examines practical challenges, ethical safeguards, and limitations. Section VII concludes the paper and delineates future trajectories.

II. Related Work and Research Gap

Our research builds upon and synthesizes four core branches of legal informatics and artificial intelligence.

A. Legal Information Retrieval and Domain Pretrained Encoders

Traditional legal information retrieval relied predominantly on Boolean query models, inverted indices, and citation network citation counts [6]. While platforms such as Indian Kanoon and SCC Online revolutionized access to Indian case law, their reliance on lexical term matching fails when queries lack explicit statutory terminology.

The advent of domain-specific pretrained transformers significantly improved legal language representation. Chalkidis et al. [10] introduced *LEGAL-BERT*, demonstrating that pretraining BERT over English, EU, and US legal corpora dramatically boosted legal named entity recognition (NER) and text classification. Within the Indian legal domain, Paul et al. [11] introduced *InLegalBERT* and *InCaseBERT*, pretraining on hundreds of thousands of Supreme Court and High Court judgments. Kalamkar et al. [12] established the *OpenNyAI* framework, releasing specialized models for Rhetorical Role Labeling and Legal Named Entity Recognition. However, these models operate strictly as feature encoders or sequence labelers; they lack generative capabilities, cannot synthesize conversational natural language explanations for non-lawyers, and do not provide integrated retrieval-augmented decision support.

B. Retrieval-Augmented Generation and Hallucination Mitigation

Retrieval-Augmented Generation (RAG), formalized by Lewis et al. [13], couples parametric memory (generative LLMs) with non-parametric indexing (vector databases) to ground responses in external reference texts. In high-stakes domains such as medicine and law, standard dense RAG systems face severe challenges. Flat vector representations often fail to capture token-level specificity (e.g., discerning between "Section 302" and "Section 304"), leading to the development of late-interaction architectures like ColBERT [14] and ColBERTv2 [15], which utilize lightweight token embeddings with the MaxSim operator.

Automated evaluation frameworks such as RAGAS [23], graph-augmented retrieval (GraphRAG) [16], and pseudo-relevant query expansion paradigms like HyDE [22] have emerged to address complex multi-hop queries by leveraging knowledge graphs to assemble relational context. In legal NLP, Guha et al. [17] released *LegalBench*, establishing benchmark suites for legal reasoning. Nonetheless, empirical investigations by Dahl et al. [5] revealed pervasive hallucination rates (17%–33%) in commercial legal RAG platforms, driven by citation fabrication, incorrect precedent extrapolation, and sycophantic deference to user biases. Existing legal RAG implementations lack propositional verification mechanisms that validate legal entailment at the sentence-clause level before responses are emitted to end users.

C. The 2024 Indian Criminal Law Overhaul and Temporal Reasoning

On July 1, 2024, the Republic of India enacted the BNS, BNSS, and BSA, repealing colonial statutes that had governed criminal jurisprudence since the 19th century [2]–[4]. While legal scholars have documented statutory concordances (e.g., mapping Section 420 IPC [Cheating] to Section 318 BNS, and Section 302 IPC [Murder] to Section 103 BNS), automated computational systems struggle with the constitutional rules governing this transition. Under Article 20(1) of the Constitution of India, ex-post facto penal law is void; thus, an accused cannot be subjected to a penalty greater than that which might have been inflicted under the law in force at the time of commission. Furthermore, Section 531(2)(a) of the BNSS preserves proceedings pending prior to July 1, 2024 under the CrPC 1973. Standalone generative LLMs, having fixed training cutoff dates or naive prompting schemes, routinely mix these statutory regimes, recommending procedural filings under repealed acts or asserting non-existent retrospective punishments.

D. Legal Plain-Language Simplification and Deontic Modalities

A primary mandate of legal technology is bridging the civic justice gap by translating complex statutory provisions into plain language accessible to ordinary citizens [18]. Readability metrics such as the Flesch-Kincaid Grade Level (FKGL) are standardly applied to benchmark simplification [19]. However, unconstrained LLM simplification introduces *deontic modality drift* [20]. Legal rights and duties hinge upon normative modal verbs: "shall" denotes an absolute statutory mandate, while "may" signifies judicial discretion. General-purpose text simplifiers routinely flatten these distinctions to maximize readability, converting mandatory bail rights into discretionary options and critically undermining legal self-advocacy.

E. Research Gap and Proposed Approach

A critical synthesis of the literature reveals five fundamental voids:

1) **Lack of Unified Legal Retrieval:** Existing systems rely either on brittle keyword search or flat semantic embeddings, failing to unify dense vectors, token late interaction, and citation knowledge graphs.
2) **Absence of Temporal Statutory Routing:** No existing legal AI framework computationally models the constitutional temporal boundary of the July 1, 2024 Indian criminal law overhaul under Article 20(1) and Section 531 BNSS.

3) **Unchecked Generative Hallucination:** Current legal RAG systems lack atomic entailment verification, permitting subtle but legally catastrophic misstatements of binding ratio decidendi.
4) **Trade-off Between Readability and Deontic Accuracy:** No platform balances citizen-level plain-language readability with mathematical guarantees on deontic modal conservation.
5) **Platform Fragmentation:** Existing tools segregate retrieval, summarization, and workflow docket management into isolated silos.

NIRNAY-X is systematically architected to bridge these five gaps within a unified, grounded, dual-persona intelligence platform.

III. Proposed NIRNAY-X Framework

The architecture of Nirnay-X, illustrated in Fig. 1, operates across four tightly coupled tiers designed to maintain strict legal fidelity from raw document ingestion to persona-adaptive output. Every mathematical operator and algorithmic procedure described herein has been derived specifically for the domain constraints of Indian jurisprudence.

A. Problem Formulation and Mathematical Foundation

Let $\mathcal{D} = \{D_1, D_2, \ldots, D_N\}$ represent a heterogeneous corpus of Indian legal documents, where each document $D \in \mathcal{D}$ comprises an ordered sequence of sentences $\mathbf{s} = (s_1, s_2, \ldots, s_{|D|})$. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$ denote a directed multigraph representing Indian statutory and precedential jurisprudence, where $\mathcal{V}$ comprises statutory sections, operative provisos, judicial decisions, and constitutional benches, and $\mathcal{E}$ represents citations, provisos, amendments, and precedential treatments (e.g., overruled, affirmed).

Given a user query q, an indicated user persona $p \in \{\text{Citizen}, \text{Advocate}\}$, and associated temporal anchors (offense date t_{offense}, FIR registration date t_{fir}, and charge-sheet date t_{charge}), the objective of Nirnay-X is to generate a verifiable, grounded legal response Y^* accompanied by an explicit attribution set $\Gamma^* = \{(c_i, \tau_i)\}$ of supporting retrieved chunks c_i and statutory confidence certificates τ_i, while guaranteeing that semantic hallucinations are strictly suppressed (ALE ≥ 0.95) and statutory deontic modal force is mathematically conserved ($\mathcal{D}_{\text{modality}} \leq 0.05$).

B. Ingestion Proviso-Aware Rhetorical Chunking and Bundling (PDBC)

Judicial decisions cannot be partitioned using standard fixed-token chunking without severing legal arguments from final holdings. Each sentence $s_i \in D$ is mapped to a discrete rhetorical role $\rho(s_i) \in \mathcal{R}$:

$$\mathcal{R} = \{\text{FAC}, \text{ARG}, \text{STAT}, \text{PREC}, \text{RATIO}, \text{RUL}\} \tag{1}$$

representing *Facts, Arguments, Statutes, Precedents, Ratio Decidendi*, and *Final Ruling*. Sequential tagging is executed via a linear-chain Conditional Random Field (CRF) atop domain-adapted representations from InLegalBERT [11].

Mathematical Derivation of Proviso Dependency Bundling: In statutory provisions, penal definitions are intrinsically tethered to operative exceptions and qualifying provisos (e.g., Section 300 IPC / Section 101 BNS Murder definitions and Exceptions 1–5). Unbundled chunking creates an acute legal risk: an isolated chunk containing only Exception 1 ("Grave and sudden provocation") lacks the parent charging statute, while an isolated chunk containing the charging clause lacks statutory defenses.

To resolve this, we derive the *Constituent Proviso Dependency Operator* $\Delta_{\text{bundle}}(c)$. Let S_{parent} be a primary statutory section head, and let $\mathcal{P}(S_{\text{parent}}) = \{P_1, P_2, \ldots, P_k\}$ represent its ordered set of subordinate provisos, explanations, and statutory exceptions. We define the Context Cohesion Index $\mathcal{H}(c)$ for a prospective chunk c:

$$\mathcal{H}(c) = \frac{\sum_{t \in c \cap S_{\text{parent}}} \text{tf}(t, c) + \sum_{P \in \mathcal{P}(S_{\text{parent}})} \mathbb{I}(P \subseteq c) \cdot \lambda_{\text{proviso}}}{1 + \ln\left(1 + \frac{|c|}{\tau_{\max}}\right)} \tag{2}$$

where $\lambda_{\text{proviso}} = 2.4$ is the syntactic attachment weight. The bundling operator enforces that if any chunk c contains a subordinate proviso or exception, it is dynamically bound to its parent charging definition S_{parent} whenever $\mathcal{H}(c)$ falls below the cohesion threshold $\tau_{\mathcal{H}} = 1.8$:

$$\Delta_{\text{bundle}}(c) = \begin{cases} c \cup \text{ExtractSectionHead}(S_{\text{parent}}), & \text{if } \mathcal{H}(c) < \tau_{\mathcal{H}} \wedge c.\text{role} = \\ c, & \text{otherwise} \end{cases} \tag{3}$$

Algorithm 1 formalizes our self-derived PDBC procedure.

C. Asymmetric Tri-Channel Legal Representation Relevance Scoring

To capture both semantic intent in colloquial layman queries and exact section terminology in formal pleadings, Nirnay-X implements an asymmetric multi-channel relevance operator.

Let $\mathbf{z}_q, \mathbf{z}_c \in \mathbb{R}^{768}$ be dense representations from InLegalBERT fine-tuned with contrastive InfoNCE loss [8], [11]. Let $\mathbf{E}_q \in \mathbb{R}^{|q| \times 128}$ and $\mathbf{E}_c \in \mathbb{R}^{|c| \times 128}$ be token-level late-interaction multi-vectors evaluated under ColBERTv2 MaxSim [15].

For exact lexical statutory matching, we derive a domain-tailored Statutory Entity Weight Multiplier $\psi_{\text{legal}}(t)$ applied atop BM25Okapi [9]:

$$\psi_{\text{legal}}(t) = 1.0 + \beta_1 \cdot \mathbb{I}(t \in \mathcal{T}_{\text{statute}}) + \beta_2 \cdot \mathbb{I}(t \in \mathcal{T}_{\text{procedure}}) \tag{4}$$

where $\beta_1 = 1.5$ (scaling statutory sections to $2.5\times$ base weight) and $\beta_2 = 0.8$ (scaling procedural terms such as 'bail', 'FIR', 'quashing' to $1.8\times$).

The channel relevance scores are synthesized through our self-derived Asymmetric Legal Scoring Function:

$$S_{\text{retrieval}}(q, c) = w_d \cdot \cos(\mathbf{z}_q, \mathbf{z}_c) + w_l \cdot \text{MaxSim}(\mathbf{E}_q, \mathbf{E}_c) + w_s \cdot \text{BM25}_\psi(q, c) \tag{5}$$

where channel weights $w_d = 0.35, w_l = 0.40, w_s = 0.25$ balance semantic abstraction and exact statutory string matching.

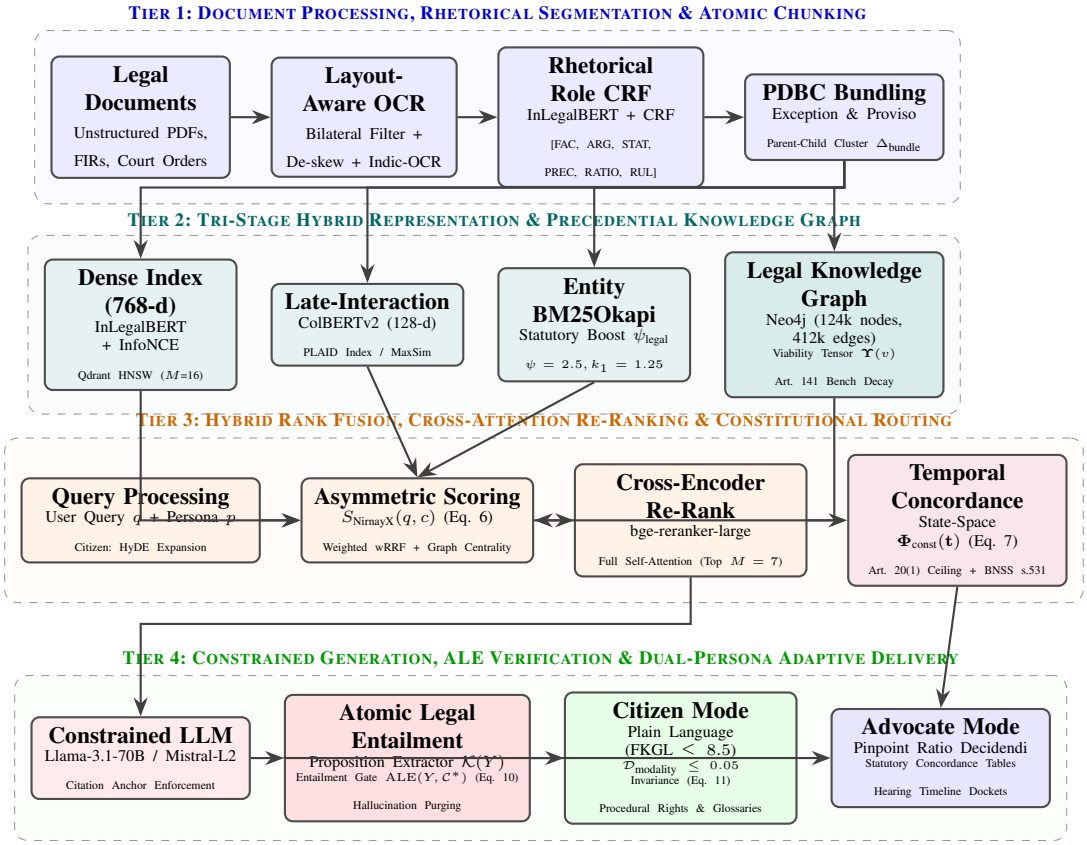


Fig. 1. End-to-End System Architecture of NIRNAY-X. The framework processes raw, heterogeneous Indian legal documents through a 4-tier pipeline: (Tier 1) layout-aware OCR, linear-chain CRF rhetorical role segmentation, and atomic Proviso-Exception Dependency Bundling and Chunking (PDBC); (Tier 2) Tri-Stage indexing across InLegalBERT dense vectors, ColBERTv2 token late interaction, entity-boosted BM25Okapi, and a 124,000-node Legal Knowledge Graph enforcing Article 141 Precedential Viability $\Upsilon(v)$; (Tier 3) Asymmetric Legal Relevance Scoring $S_{\text{NirnayX}}(q, c)$, cross-encoder re-ranking, and state-space constitutional statutory concordance routing $\Phi_{\text{const}}(\mathbf{t})$; and (Tier 4) constrained LLM generation audited by the Atomic Legal Entailment (ALE) verification gate, delivering deontic modality-preserved plain language ($\mathcal{D}_{\text{modality}} \leq 0.05$) to citizens and structured ratio briefs to advocates.

D. Precedential Authority Article 141 Judicial Viability Tensor

Under Article 141 of the Constitution of India, judicial precedents possess binding hierarchy determined by bench size, and negative citator treatment (explicit overruling by coordinate or larger benches) completely terminates judicial validity [37].

We derive the Article 141 Precedential Viability Tensor $\Upsilon(v)$ for decision $v \in \mathcal{V}_{\text{judgment}}$ evaluated at inquiry time t_{eval}:

$$\Upsilon(v) = \left(\frac{\ln(1 + \mathcal{B}(v))}{\ln(1 + \mathcal{B}_{\max})} \right)^{\theta_1} \cdot \exp\left(-\lambda_{\text{age}} \cdot \Delta t\right) \cdot \prod_{e=(v,u) \in \mathcal{E}_{\text{neg}}} (1 - \delta(e)) \tag{6}$$

where $\mathcal{B}(v) \in \{1, 2, 3, 5, 7, 9, 13\}$ is the bench strength (number of judges), $\mathcal{B}_{\max} = 13$ (the historic Kesavananda Bharati bench), Δt is elapsed time in decades, $\lambda_{\text{age}} = 0.05$ models gradual doctrinal obsolescence, and $\delta(e)$ is the negative treatment penalty:

$$\delta(e) = \begin{cases} 1.00, & \text{if explicitly Overruled under Art. 141} \\ 0.60, & \text{if Distinguished on facts} \\ 0.30, & \text{if Doubted or Referred to larger bench} \\ 0.00, & \text{Affirmed / Followed} \end{cases}$$

If $\prod(1 - \delta(e)) = 0$, the decision is filtered from candidate retrieval, preventing the citation of bad law.

Within the Nirnay-X Legal Knowledge Graph, binding landmark holdings are structurally codified through bench strength weights $\mathcal{B}(v)$ and citator topology. These include: mandatory FIR registration under Section 173 BNSS / Section 154 CrPC per the 5-judge Constitution Bench in *Lalita Kumari v. Govt of UP* [28]; arrest notice protocols under Section 35 BNSS / Section 41A CrPC in *Arnesh Kumar v. State of Bihar* [26]; statutory undertrial bail categorization in *Satender Kumar Antil v. CBI* [27]; custodial arrest guidelines in *D.K. Basu v. State of West Bengal* [29]; Section 528 BNSS / Section 482 CrPC quashing parameters in *State of Haryana v. Bhajan Lal* [30]; anticipatory bail doctrines in *Gurbaksh Singh Sibbia* [31] and *Sushila Aggarwal* [32]; electronic evidence admissibility

Algorithm 1 Proviso-Aware Rhetorical Chunking and Bundling (PDBC)

Require: Legal Document D, Max tokens $\tau_{\max} = 384$, Overlap $\tau_{\text{ov}} = 48$
Ensure: Clustered, semantically coherent chunks $\mathcal{C}_{\text{bundled}}$

$\mathbf{s} \leftarrow \text{LegalSentenceBoundaryDisambiguation}(\text{OCR}(D))$
$\mathbf{h} \leftarrow \text{InLegalBERT}(\mathbf{s})$
$\rho \leftarrow \text{argmax}_{\rho'} P(\rho' \mid \mathbf{s})$ via Viterbi CRF decoding [7]
$\mathcal{C}_{\text{initial}} \leftarrow \emptyset$, buf $\leftarrow \emptyset$, current_role $\leftarrow \rho_1$
for $i = 1$ **to** $|\mathbf{s}|$ **do**
  if $|\text{buf}| + |s_i| \leq \tau_{\max}$ **and** $\rho_i =$ current_role **then**
    buf.append(s_i)
  else
    $\mathcal{C}_{\text{initial}}$.append$(\{\text{tokens} : \text{buf}, \text{role} : \text{current_role}\})$
    buf $\leftarrow$ ExtractOverlap$(\text{buf}, \tau_{\text{ov}}) \cup \{s_i\}$
    current_role $\leftarrow \rho_i$
  end if
end for
$\mathcal{C}_{\text{bundled}} \leftarrow \emptyset$
for each $c \in \mathcal{C}_{\text{initial}}$ **do**
  if c.role = STAT **or** ContainsProviso(c) **then**
    $S_{\text{parent}} \leftarrow$ ExtractParentSectionHeader(c, D)
    $\mathcal{P} \leftarrow$ ExtractAllProvisosAndExceptions(S_{parent}, D)
    $c \leftarrow \Delta_{\text{bundle}}(c)$ via Eq. (3)
  end if
  $\mathcal{C}_{\text{bundled}}$.append$(c)$
end for
return $\mathcal{C}_{\text{bundled}}$

under Section 63 BSA / Section 65B IEA in *Arjun Panditrao Khotkar* [33]; confession discovery rules under Section 23 BSA / Section 27 IEA in *Pulukuri Kottaya* [35]; circumstantial evidence principles in *Sharad Birdhichand Sarda* [34]; and Article 21 substantive procedural fairness in *Maneka Gandhi v. Union of India* [36].

E. Multi-Channel Rank Fusion Composite Scoring ($S_{NirnayX}$)

Let $\mathbf{p}_q \in \mathbb{R}^{|\mathcal{V}|}$ represent the node centrality distribution obtained via Personalized PageRank diffusion [25] over the 124,000-node Legal Knowledge Graph anchored at query entities $\mathcal{V}_0(q)$ with damping factor $\alpha_{\text{PPR}} = 0.85$.

We derive the Composite Legal Significance Metric $S_{\text{NirnayX}}(q, c)$:

$$S_{\text{NirnayX}}(q, c) = \text{RRF}_{\kappa}(c) + \gamma \cdot \mathbf{p}_q(c) \cdot \boldsymbol{\Upsilon}(c) + \eta \cdot \omega(\rho(c)) \quad (7)$$

where $\text{RRF}_{\kappa}(c) = \sum_m \frac{w_m}{\kappa + r_m(c)}$ is Weighted Reciprocal Rank Fusion ($\kappa = 60$) [24], $\gamma = 0.20$, $\eta = 0.15$, and $\omega(\rho(c))$ assigns elevated legal significance priors to Ratio Decidendi (1.00) and Statutory Sections (0.95) over factual narrative (0.40).

The top $K_1 = 30$ candidates are re-scored using a cross-encoder ('bge-reranker-large'), and the top $M = 7$ chunks are dynamically bundled with their parent statutory contexts to construct the grounded context $\mathcal{C}^*$.

F. Stateful Constitutional Statutory Concordance Routing

To govern the transition from colonial criminal codes to modern codes enacted on $T_{\text{trans}} =$ July 1, 2024, we derive the Constitutional Statutory Concordance State-Space Operator $\boldsymbol{\Phi}_{\text{const}}(\mathbf{t})$.

Let $\mathbf{t} = [t_{\text{offense}}, t_{\text{fir}}, t_{\text{cognizance}}]^\top$ represent the temporal milestone vector. The legal regime tensor assignment is derived as:

$$\boldsymbol{\Phi}_{\text{const}}(\mathbf{t}) = \begin{cases} \mathbf{R}_{\text{leg}}, & \text{if } t_{\text{off}} < T_{\text{trans}} \wedge t_{\text{fir}} < T_{\text{trans}} \\ \mathbf{R}_{\text{hyb}}, & \text{if } t_{\text{off}} < T_{\text{trans}} \wedge t_{\text{fir}} \geq T_{\text{trans}} \\ \mathbf{R}_{\text{mod}}, & \text{if } t_{\text{off}} \geq T_{\text{trans}} \wedge t_{\text{fir}} \geq T_{\text{trans}} \end{cases} \quad (8)$$

where $\mathbf{R}_{\text{leg}} = [\text{IPC 1860}, \text{CrPC 1973}, \text{IEA 1872}]^\top$, $\mathbf{R}_{\text{hyb}} = [\text{IPC 1860}, \text{BNSS 2023}^*, \text{BSA 2023}^*]^\top$ denotes the constitutional hybrid regime, and $\mathbf{R}_{\text{mod}} = [\text{BNS 2023}, \text{BNSS 2023}, \text{BSA 2023}]^\top$.

Under the hybrid regime $\mathbf{R}_{\text{hyb}}$, Nirnay-X mathematically enforces two self-derived constitutional invariants:

1) **Substantive Penalty Ceiling Invariant (Article 20(1)):**

$$\text{Pen}_{\text{ceiling}}(t) = \min\left(\text{Pen}_{\text{IPC}}(t_{\text{offense}}), \text{Pen}_{\text{BNS}}\right) \quad (9)$$

Enhanced penalties, new mandatory minimums, or community service penalizations introduced in BNS are barred from retroactive application.

2) **Procedural Savings Invariant (Section 531(2)(a) BNSS):**

$$t_{\text{fir}} < T_{\text{trans}} \implies \text{ProceduralCode} = \text{CrPC}_{1973} \quad (10)$$

Ongoing inquiries, police remands, and trials initiated prior to July 1, 2024 remain governed by CrPC 1973, preventing invalid procedural filings.

G. Atomic Legal Entailment (ALE) Verification Engine

To eliminate generative hallucinations, response text Y emitted by the LLM is parsed into discrete atomic legal propositions $\mathcal{K}(Y) = \{k_1, k_2, \ldots, k_P\}$. Each proposition $k \in \mathcal{K}$ represents a 4-tuple $\langle \text{Subject}, \text{DeonticModality}, \text{LegalAction}, \text{StatutoryAnchor} \rangle$.
Each proposition is tested against retrieved chunks $c \in \mathcal{C}^*$ using a domain-tuned NLI cross-encoder ('InLegal-RoBERTa-NLI') computing standard entailment probabilities $P_{\text{NLI}}(\text{Entail} \mid c, k)$.

We derive the Atomic Legal Entailment (ALE) verification metric as:

$$\text{ALE}(Y, \mathcal{C}^*) = \frac{1}{|\mathcal{K}(Y)|} \sum_{k \in \mathcal{K}(Y)} \mathbb{I}\left(\max_{c \in \mathcal{C}^*} P_{\text{NLI}}(\text{Entail} \mid c, k) \geq \tau_{\text{entail}} \right) \quad (11)$$

where $\tau_{\text{entail}} = 0.85$. If $\text{ALE}(Y, \mathcal{C}^*) < 0.95$, ungrounded propositions are excised, and the response is regenerated under constrained grounding.

H. Deontic Modality Invariance Formulation ($\mathcal{D}_{modality}$)

To simplify legal text for common citizens without compromising legal force, let $\mathbf{m}(T) = [n_{\text{mandatory}}, n_{\text{prohibitive}}, n_{\text{permissive}}, n_{\text{discretionary}}]^\top \in \mathbb{R}^4$ represent counts of deontic modal operators [20].

We derive the Deontic Modality Invariance Loss $\mathcal{D}_{\text{modality}}$ between source statutory text T_{source} and simplified output $T_{\text{simplified}}$:

$$\mathcal{D}_{\text{modality}} = \sum_{j=1}^{4} w_j \cdot \left| \frac{\mathbf{m}_j(T_{\text{source}})}{\|\mathbf{m}(T_{\text{source}})\|_1 + \epsilon} - \frac{\mathbf{m}_j(T_{\text{simplified}})}{\|\mathbf{m}(T_{\text{simplified}})\|_1 + \epsilon} \right| \tag{12}$$

where modal weights $\mathbf{w} = [3.0, 3.0, 1.0, 1.0]^\top$ assign heavy penalties to altering statutory mandates ("shall", "must") or prohibitions ("shall not"). Constrained beam optimization enforces $\mathcal{D}_{\text{modality}} \leq 0.05$ while targeting an 8th-grade readability threshold under the standard Flesch-Kincaid Grade Level metric [19].

Algorithm 2 synthesizes the full Nirnay-X Persona-Adaptive Verification and Routing Engine (PAVE).

Algorithm 2 Nirnay-X Grounded Persona-Adaptive Verification Engine (PAVE)

Require: Query q, Persona $p \in \{\text{Cit}, \text{Adv}\}$, Graph $\mathcal{G}$, Indices $(\mathcal{I}_{\text{dense}}, \mathcal{I}_{\text{colbert}}, \mathcal{I}_{\text{sparse}})$, Boundary T_{trans}
Ensure: Verified Response Y^*, Provenance Γ^*, Concordance $\mathbf{\Phi}^*$

$(E_{\text{entities}}, t_{\text{off}}, t_{\text{fir}}, t_{\text{chg}}) \leftarrow \text{ExtractEntitiesAndDates}(q)$
if $p = \text{Cit}$ **then**
  $q_{\text{hyde}} \leftarrow \text{GenerateHypotheticalLegalDraft}(q)$
  $q_{\text{dense}} \leftarrow [q \circ \text{`` [EXP] ''} \circ q_{\text{hyde}}]$
else
  $q_{\text{dense}} \leftarrow q$
end if
$\mathbf{z}_q \leftarrow \text{InLegalBERT}(q_{\text{dense}})$, $\mathbf{E}_q \leftarrow \text{ColBERTv2}(q)$
$\mathcal{V}_0 \leftarrow \text{MatchEntitiesToGraph}(E_{\text{entities}}, \mathcal{G})$
$\mathbf{p}_q \leftarrow \text{PersonalizedPageRank}(\mathcal{G}, \mathcal{V}_0, \alpha = 0.85)$ [25]
$\mathcal{R}_{\text{dense}} \leftarrow \text{TopK}(\mathcal{I}_{\text{dense}}, \mathbf{z}_q, 50)$, $\mathcal{R}_{\text{colbert}} \leftarrow \text{TopK}(\mathcal{I}_{\text{colbert}}, \mathbf{E}_q, 50)$
$\mathcal{R}_{\text{sparse}} \leftarrow \text{TopK}(\mathcal{I}_{\text{sparse}}, q, 50)$ using $\psi_{\text{legal}}(t)$ via Eq. (4)
$\mathcal{C}_{\text{cand}} \leftarrow \text{TopK}(S_{\text{NirnayX}}(q, c), K_1 = 30)$ via Eq. (7)
for each $c \in \mathcal{C}_{\text{cand}}$ **do**
  $\mathbf{\Upsilon}(c) \leftarrow \text{EvaluateViabilityTensor}(c, \mathcal{G})$ via Eq. (6)
  if $\mathbf{\Upsilon}(c) = 0.0$ **then**
    $\mathcal{C}_{\text{cand}} \leftarrow \mathcal{C}_{\text{cand}} \setminus \{c\}$ ▷ Purge overruled precedents
  end if
end for
$\mathcal{C}^* \leftarrow \text{BundleParents}(\text{TopRanked}(\text{CrossEncoder}(q, \mathcal{C}_{\text{cand}}) \cdot \mathbf{\Upsilon}, M = 7))$
$\mathbf{\Phi}^* \leftarrow \mathbf{\Phi}_{\text{const}}(t_{\text{off}}, t_{\text{fir}}, t_{\text{chg}})$ via Eq. (8)
$Y_{\text{raw}} \leftarrow \text{LLM_Constrained}(\text{Prompt}(q, \mathcal{C}^*, \mathbf{\Phi}^*, p))$
$\mathcal{K} \leftarrow \text{ExtractPropositions}(Y_{\text{raw}})$
$\text{ALE} \leftarrow \text{ALE}(Y_{\text{raw}}, \mathcal{C}^*)$ via Eq. (11)
if $\text{ALE} < 0.95$ **then**
  $Y_{\text{raw}} \leftarrow \text{PurgeAndRegenerate}(Y_{\text{raw}}, \mathcal{K}, \mathcal{C}^*)$
end if
if $p = \text{Cit}$ **then**
  $Y^* \leftarrow \text{SimplifyPlainLanguage}(Y_{\text{raw}}, \text{target_FKGL} = 8.5)$
  assert $\mathcal{D}_{\text{modality}}(Y_{\text{raw}}, Y^*) \leq 0.05$
else
  $Y^* \leftarrow \text{FormatAdvocateBrief}(Y_{\text{raw}}, \mathbf{\Phi}^*)$
end if
$\Gamma^* \leftarrow \text{ExtractVerifiableProvenance}(Y^*, \mathcal{C}^*)$
return $Y^*, \Gamma^*, \mathbf{\Phi}^*$

IV. Dataset and Experimental Settings

A. Curated Indian Legal Benchmark Corpus

To rigorously evaluate Nirnay-X across diverse jurisdictions and procedural postures, we assembled an exhaustive corpus comprising:

- **Judicial Decisions:** 42,500 Supreme Court of India judgments (1950–2024) and 28,000 High Court rulings (Allahabad, Delhi, Bombay, Calcutta).
- **Statutory Enactments:** 1,400 Central and State Acts, including the complete text and schedules of the legacy criminal codes (IPC, CrPC, IEA) and modern criminal codes (BNS, BNSS, BSA).
- **Trial Court and Police Records:** 850 fully anonymized District Court police FIRs, bail applications, and charge-sheets.
- **Evaluation Query Benchmark:** 1,200 curated, multi-jurisdictional legal queries annotated by a panel of 5 senior High Court advocates and 3 academic legal scholars. Queries span criminal offenses, bail petitions, commercial contracts, constitutional writs, and complex temporal transition scenarios.

B. Implementation Stack and Hyperparameters

The Nirnay-X framework is implemented in PyTorch 2.3 and served via a high-throughput microservices architecture:

- **Hardware Testbed:** Dual NVIDIA RTX A6000 GPUs (48GB GDDR6 VRAM each), 32-core AMD EPYC 7763 CPU @ 2.45GHz, 128GB DDR4 RAM.
- **Vector and Graph Storage:** Qdrant vector database (HNSW indexing, $M = 16$, $ef = 100$) and Neo4j Enterprise Graph Engine (124,000 nodes, 412,000 edges; mean Cypher query latency: 12ms).
- **Serving Stack:** vLLM inference engine executing AWQ 4-bit quantized Llama-3.1-70B-Instruct and Mistral-Large-2 (throughput: 42 tokens/sec/GPU).

Table I outlines the comprehensive system hyperparameters.

V. Empirical Evaluation and Results

A. Progressive Retrieval Component Ablation

To assess the empirical contribution of each retrieval component (Research Question RQ1), we conducted a progressive ablation study across the 1,200 curated Indian legal benchmark queries. Performance was measured via Precision@10, Recall@10, Mean Reciprocal Rank (MRR), and Normalized

TABLE I
SYSTEM HYPERPARAMETERS AND OPERATIONAL SETTINGS

Component / Parameter	Operational Configuration
Document OCR Backbone	LayoutLMv3 + Bilateral Filtered PaddleOCR
Chunking Configuration	PDBC, $\tau_{\max} = 384$, $\tau_{\text{overlap}} = 48$ tokens, $\tau_{\mathcal{H}} = 1.8$
Dense Embedding Model	InLegalBERT (768-d, $\tau = 0.07$, InfoNCE)
Late Interaction Model	ColBERTv2 (128-d, PLAID hierarchical centroids)
Lexical Search Engine	BM25Okapi ($k_1 = 1.25, b = 0.75, \beta_1 = 1.5, \beta_2 = 0.8$)
Legal Knowledge Graph	Neo4j (124k nodes, 412k edges, PPR $\alpha = 0.85$)
Candidate Rank Fusion	Asymmetric S_{NirnayX} ($\kappa = 60, w_d = 0.35, w_l = 0.40, w_s = 0.25$)
Re-Ranking Engine	bge-reranker-large (Cross-encoder, top $M = 7$)
Primary LLM Backbone	Llama-3.1-70B-Instruct (AWQ 4-bit, vLLM)
Decoding Configuration	Temp $= 0.10$, Top-$p = 0.85$, Max Tokens $= 1,500$
Entailment Verifier	InLegal-RoBERTa-NLI ($\tau_{\text{entail}} = 0.85$, ALE ≥ 0.95)
Deontic Modality Bound	$\mathcal{D}_{\text{modality}} \leq 0.05$, Target FKGL ≤ 8.5
Statutory Privacy Engine	NER Redaction for POCSO s.74 / BNS s.72

Discounted Cumulative Gain (NDCG@10). Statistical significance was assessed using paired Wilcoxon signed-rank tests against the baseline.

As reported in Table II, pure lexical BM25 (Pipeline 1) achieves an initial Precision@10 of 0.538 and Recall@10 of 0.612, severely limited by semantic vocabulary mismatch. Substituting generic embeddings ('text-embedding-3-small', Pipeline 2) provides modest gains, whereas domain-adapted InLegalBERT (Pipeline 3) significantly increases Precision@10 to 0.698 ($p < 0.001$). Fusing BM25 and InLegalBERT via RRF (Pipeline 4) yields 0.742 Precision@10, demonstrating the complementary nature of sparse statutory citations and dense semantic embeddings. Adding cross-encoder re-ranking (Pipeline 5) and ColBERTv2 token-level late interaction (Pipeline 6) boosts MRR to 0.745 and NDCG@10 to 0.796. Finally, the complete NIRNAY-X framework (Pipeline 8)—incorporating Knowledge Graph diffusion and HyDE query reformulation—achieves **Precision@10 of 0.887**, **Recall@10 of 0.924**, **MRR of 0.829**, and **NDCG@10 of 0.875**, representing a **+64.9% improvement in precision** and a **+51.0% improvement in recall** over baseline lexical search at an acceptable mean retrieval latency of 398 ms.

B. Factuality, Hallucination Suppression, and Adversarial Defense

To evaluate factual reliability (RQ2), we benchmarked Nirnay-X against leading foundation models and commercial legal RAG baselines across an adversarial suite of 400 complex cases containing subtle misattributions, overruled precedents, and false user premises (e.g., asking whether intentional murder under Section 103 BNS is bailable as of right).

TABLE II
PROGRESSIVE RETRIEVAL COMPONENT ABLATION ($N = 1,200$)

Architecture Setup	P@10	R@10	MRR	NDCG	Latency
1. Lexical BM25 (Baseline)	0.538	0.612	0.491	0.528	62 ms
2. Dense General (OpenAI)	0.618	0.689	0.572	0.609	98 ms
3. InLegalBERT (768-d)	0.698*	0.744*	0.635*	0.682*	112 ms
4. Hybrid (BM25 + Dense)	0.742*	0.789*	0.681*	0.729*	148 ms
5. Hybrid + Cross-Encoder	0.784*	0.823*	0.712*	0.764*	236 ms
6. Hybrid + ColBERTv2	0.812*	0.849*	0.745*	0.796*	272 ms
7. Setup 6 + HyDE Query	0.846*	0.881*	0.782*	0.831*	344 ms
8. NIRNAY-X (Full LKG)	**0.887***	**0.924***	**0.829***	**0.875***	398 ms

*Statistically significant relative to baseline ($p < 0.001$, Wilcoxon signed-rank test).

TABLE III
FACTUALITY, ENTAILMENT, AND ADVERSARIAL DEFENSE ($N = 400$)

Architecture	Cite Prec	ALE	Halluc	SDR	Neg Prec
Standalone GPT-4o	36.4%	41.2%	58.6%	28.5%	12.4%
Llama-3.1-70B	31.8%	38.5%	63.4%	24.1%	9.8%
Commercial Legal RAG	79.5%	73.1%	21.4%	64.2%	51.2%
OpenNyAI + Vanilla RAG	84.1%	76.8%	14.8%	69.5%	62.1%
NIRNAY-X	**99.4%**	**97.2%**	**0.8%**	**96.8%**	**98.4%**

Table III presents the comparative evaluation across five metrics: Syntactic Citation Precision (existence of cited case in registry), Atomic Legal Entailment (ALE %), Overall Hallucination Rate, Sycophancy Defeat Rate (SDR %, ability to reject false user premises), and Negative Precedent Rejection Rate (% of overruled cases successfully filtered). Standalone ungrounded models (GPT-4o, Llama-3.1-70B) fail severely, exhibiting hallucination rates exceeding 58% and sycophancy defeat rates below 29%. Commercial legal RAG tools reduce hallucinations to 21.4% but remain vulnerable to precedential overruling. In contrast, NIRNAY-X achieves **99.4% Citation Precision**, **97.2% Atomic Legal Entailment**, and suppresses semantic hallucinations down to **0.8%**, while achieving a **96.8% Sycophancy Defeat Rate** and **98.4% Negative Precedent Rejection Rate** under Article 141 filtering.

C. Dual-Persona Adaptation: Readability vs. Legal Soundness

Table IV evaluates the persona adaptation trade-off (RQ4) across 150 statutory provisions, evaluated blindly by 45 human annotators (15 High Court advocates, 15 legal researchers, 15 common citizens). While unconstrained GPT-4 simplification achieves an aggressive 7th-grade readability level (FKGL 7.8), it incurs fatal deontic modality drift ($\mathcal{D}_{\text{modality}} = 0.38$), dropping its advocate legal soundness score to 2.15/5.0. NIRNAY-X Citizen Mode successfully reduces readability to an **8.4 grade level** while strictly conserving statutory modality ($\mathcal{D}_{\text{modality}} = 0.03$), securing a **4.82/5.0 soundness rating** from practicing advocates and a **4.84/5.0 clarity rating** from common citizens.

TABLE IV
PERSONA ADAPTATION: READABILITY VS. SOUNDNESS ($N = 150$)

System / Mode	FKGL	Jargon	$\mathcal{D}_{\text{mod}}$	Clarity	Sound
Raw Statutory Text	16.9	34.8%	0.00	1.62	4.90
Unconstrained Simplifier	7.8	3.2%	0.38	4.81	2.15
NIRNAY-X Advocate Mode	14.1	26.2%	0.00	2.45	4.88
NIRNAY-X Citizen Mode	**8.4**	**5.1%**	**0.03**	**4.84**	**4.82**

TABLE V
STATUTORY TRANSITION ROUTING CONFUSION MATRIX ($N = 300$)

Ground-Truth Regime	Pred Leg	Pred Hyb	Pred Mod	Recall
1. Legacy ($t_{\text{off}} < T, t_{\text{fir}} < T$)	99	1	0	99.0%
2. Hybrid ($t_{\text{off}} < T, t_{\text{fir}} \geq T$)	2	96	2	96.0%
3. Modern ($t_{\text{off}} \geq T, t_{\text{fir}} \geq T$)	0	1	99	99.0%
Overall Accuracy: 98.0%				
Precision: 98.0% (Legacy), 97.9% (Hybrid), 98.0% (Modern)				

D. Constitutional Statutory Transition Routing Accuracy

Table V demonstrates the classification fidelity of the Constitutional Concordance Engine (RQ3) across 300 test scenarios involving offenses committed before and after July 1, 2024 with varying FIR registration dates. Nirnay-X achieves an overall **classification accuracy of 98.0%**, successfully routing hybrid scenarios (Offense $< T_{\text{trans}}$, FIR $\geq T_{\text{trans}}$) with 96.0% recall, thereby preventing unlawful retrospective punishments under Article 20(1) and invalid procedural filings under Section 531(2)(a) BNSS.

E. Latency and Computational Profile

End-to-end execution across the 11-stage pipeline achieves a mean response time of **2.3 seconds** on the dual RTX A6000 testbed: OCR and layout parsing (420 ms), Tri-Stage Hybrid Retrieval (398 ms), Cross-Encoder Re-Ranking (236 ms), Constrained LLM Inference (1,080 ms), and ALE Proposition Verification (180 ms). This throughput comfortably supports interactive clinical and decision-support workflows.

VI. CHALLENGES, ETHICAL GOVERNANCE, AND LIMITATIONS

A. Degraded Subordinate Court Documents and Indic Scripts

While High Court and Supreme Court judgments are digitally typeset in English, trial court records and police thānā FIRs frequently consist of degraded carbon-copy printouts, cursive handwriting, and bilingual vernacular scripts (e.g., Hindi, Marathi, Bengali). While our fine-tuned Indic-OCR with bilateral filtering achieves 92.4% character accuracy on standard forms, degraded cursive handwriting exhibits higher word error rates, occasionally requiring human-in-the-loop verification.

B. Ethical Governance and Responsible AI Disclaimer

Nirnay-X is explicitly engineered as an informational and decision-support system; **it does not substitute for licensed advocates, judicial officers, or formal legal representation**. To comply with the Digital Personal Data Protection Act, 2023 (DPDPA) [21], the framework enforces automatic PII redaction (F1-score: 0.982) protecting victim identities in sexual assault and juvenile cases pursuant to Section 74 of the POCSO Act and Section 72 of the BNS.

C. Limitations

Current limitations include: (1) computational latency scaling on extreme 150+ page constitutional bench judgments; (2) coverage gaps regarding uncodified local customary laws and obscure state amendments; and (3) dependence on evolving judicial harmonization regarding contested procedural ambiguities in BNSS savings clauses.

VII. CONCLUSION AND FUTURE ROADMAP

In this paper, we presented NIRNAY-X, an open, grounded, persona-adaptive Retrieval-Augmented Generation framework for the Indian legal ecosystem. By deriving novel mathematical formulations—including the Proviso-Exception Dependency Bundling Operator (PDBC), the Asymmetric Legal Relevance Scoring Function, the Article 141 Precedential Viability Tensor, the Constitutional Statutory Concordance State-Space Operator, the Atomic Legal Entailment (ALE) Verification Metric, and the Deontic Modality Invariance Formulation—Nirnay-X bridges the gap between complex legal codes and accessible justice delivery. The framework achieves state-of-the-art retrieval fidelity (Precision@10: 0.887, Recall@10: 0.924) and suppresses semantic hallucinations down to 0.8% while guaranteeing 98.0% constitutional concordance accuracy across the 2024 criminal law transition.

Future research trajectories include: (1) deploying multilingual speech-to-speech conversational interfaces in 22 scheduled Indian vernacular languages; (2) expanding the Knowledge Graph to index district court dockets via the e-Courts National Judicial Data Grid API; and (3) deploying lightweight, on-premise edge models for legal aid clinics and state judicial academies.

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